# SURAJ JAYBHAYE

FULL STACK DEVELOPER | AI ENGINEER
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## SUMMARY

Full stack developer with **1+ year** of experience building production web applications and **AI-powered products** using **Next.js, TypeScript, Node.js** and **Python**. **Computer Engineering** graduate from **COEP Technological University** with a strong foundation in **data structures and algorithms** (**250+ problems solved**) and hands-on experience in **RAG pipelines, LLM integrations** and **AWS**. Shipped a **multi-tenant ERP** serving **20+ branches** and RAG search that cut policy lookups from ~30 minutes to under 10 seconds.

## EXPERIENCE

**Invade Code Ltd.** **Pune, India**
*Full Stack Developer* *Jan 2026 – Present*

**IAG ERP** | *Next.js, TypeScript, Firebase, Google Genkit*

- Built a role-based agro-retail **ERP** on **Firebase** (Firestore, Auth, Cloud Functions), delivering **95+ pages** across POS, inventory, procurement, branch transfers, finance and HR for **20+ branches** and **13 user roles**.
- Developed a **POS billing module** handling **1,000+ bills daily**, with batch- and expiry-aware stock selection, loose-unit (g/ml) pricing, UPI payment proof, **QR-linked digital bills** and cash drawer reconciliation, using **Firestore transactions** for consistent returns.
- Implemented **GST compliance tooling** including HSN summaries, **GSTR-2B reconciliation**, multi-GSTIN support and portal-ready JSON export, with Excel and PDF reporting across 26 modules.
- Integrated **AI features** using **Google Genkit and Gemini 2.5 Flash**: invoice data extraction, anomaly flagging, cash-flow forecasting and an **ERP copilot** that answers questions from live Firestore data.

**Legal OS** | *Next.js, TypeScript, Firebase, Firestore Security Rules*

- Built a **self-service timesheet correction workflow** with approver sign-off, a **leave-management module** and native **.docx monthly reports** for a multi-tenant legal practice platform serving **9 user roles**.
- Fixed a **timer concurrency bug** that double-counted and dropped logged hours by resolving a **React effect-cleanup race** against a live Firestore listener, corrected **250+ affected time records**, and kept data consistent across 3 collections with **atomic batch writes**.
- Cut **Firestore read costs** after opening the timesheet to all staff by scoping each user's live queries to their own records instead of loading the whole firm's data.
- Resolved firm-wide **permission-denied errors** by rewriting **Firestore security rules** to satisfy Firestore's list-query validation constraints.

**GovIntel** | *Python, FastAPI, Gemini, Qdrant, Elasticsearch*

- Co-built (2-member team) an **AI policy-search platform**, a **POC for Kerala State IT Mission**, letting officers query hundreds of G.O.s, circulars and acts in **English and Malayalam** with cited answers, AI-drafted orders and summaries, cutting policy lookup from **~30 minutes** of manual search (officer-reported) to **under 10 seconds**.
- Engineered the **hybrid RAG retrieval pipeline**, combining **Qdrant** vector search (384-dim multilingual-e5 embeddings) with **Elasticsearch BM25** and merging results via **Reciprocal Rank Fusion**, to handle both semantic queries and exact G.O. number lookups.
- Built the document ingestion pipeline with **Gemini-based OCR** for scanned PDFs, ~512-word chunking and automatic metadata extraction (department, G.O. number, date), making new documents searchable within minutes.

**LinkedIn Job-Posting Automation** | *n8n, Gemini, Gmail API, LinkedIn API*

- Built an **AI-powered n8n workflow** that reads incoming job-request emails, uses **Gemini** to extract role details and generate post copy, and publishes to **LinkedIn** automatically with no manual steps.
- Handles **70–80 job posts weekly**, cutting posting time from **20–30 minutes to under 1 minute** per post and saving **20+ hours of manual work every week**.

**Nebula Technology** **Pune, India**
*Software Developer Intern* *Sep 2024 – Mar 2025*

**Banking Web Platform** | *React, Node.js, Express, PostgreSQL*

- Developed **customer-facing banking pages** in **React** for a live bank client, including the **account dashboard, transaction history and fund transfer** flows.

- Built and maintained **REST APIs** in **Node.js and Express** powering account, transaction and transfer operations on a **PostgreSQL** database.

## PROJECTS

**UPI Offline Mesh** | *Java, Spring Boot, JUnit* GitHub

- Built a **Spring Boot** backend for offline UPI-style payments that relays transactions through a **simulated Bluetooth mesh** and settles them once a bridge node regains connectivity.
- Secured payment packets with hybrid **RSA-OAEP + AES-256-GCM encryption** and a **two-layer replay defense** (24-hour freshness window plus per-packet nonce).
- Guaranteed **exactly-once settlement** under concurrent duplicate delivery, verified with **JUnit concurrency tests**.

## SKILLS

**Languages:** TypeScript, JavaScript, Python, Java, SQL

**Frontend:** React, Next.js, Tailwind CSS, shadcn/ui

**Backend:** Node.js, Express, FastAPI, Spring Boot, REST APIs

**AI & Automation:** RAG, LLM Integration (Gemini), Google Genkit, Qdrant, Elasticsearch, Embeddings, OCR, n8n

**Databases & Cloud:** Firebase (Firestore, Auth, Cloud Functions, App Hosting), Google Cloud, AWS (EC2, S3, Lambda), PostgreSQL, MySQL, Docker

**Tools:** Git, GitHub, JUnit, Claude Code

## EDUCATION

**COEP Technological University** **Pune, India**
*B.Tech, Computer Engineering* *2020 – 2024*

- Ranked **103rd** in **MHT-CET** (99.92 percentile) among **540,000+** candidates in Maharashtra.

**Residential Junior College** **Ahilyanagar, India**
*HSC, Science* *2018 – 2020*

- Scored **93.23%** and secured **1st Rank in Ahilyanagar District** in the 12th HSC Board Examination.